

# Exact Organization of Density Matrices and Entanglement Structure in the Kitaev Spin Liquid

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# Outline

- 1 Motivation
- 2 Kitaev Model
- 3 Organization: Equivalence Relation
- 4 Organization: Block-Diagonal Structure
- 5 Review: W.-T. Xu *et al.* PRX **15**, 011001 (2025)
- 6 Block-Diagonal Density Matrix in the Kitaev Model
- 7 Odd Number of Sites
- 8 Summary
- 9 Unsolved Problems and Outlooks

# Motivation

- density matrix
- entanglement
- spin representation to understand everything (gauge, matter, entanglement...)
- W.-T. Xu *et al.* PRX **15**, 011001 (2025)

# Kitaev Model: an Introduction

- Model Hamiltonian

$$H = - \sum_{\langle ij \rangle} J_{\alpha_{ij}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \quad ; \quad \begin{array}{c} \text{Diagram of three adjacent hexagons sharing a central vertex. The central vertex is labeled } z. \text{ The two vertices adjacent to } z \text{ are labeled } y \text{ and } x. \end{array}$$

- Majorana fermion (parton) representation:  $b^x, b^y, b^z, c \Rightarrow \sigma^\mu = ib^\mu c$
- Parton Hamiltonian: define  $ib_i^{\alpha_{ij}} b_j^{\alpha_{ij}} = \hat{u}_{ij}$

$$H \rightarrow \sum_{\langle ij \rangle} i J_{\alpha_{ij}} \hat{u}_{ij} c_i c_j$$

- Gutzwiller projection

$$\mathcal{P}_G = \prod_i \left( \frac{1 + D_i}{2} \right) \quad , \quad D_i = b_i^x b_i^y b_i^z c_i \leftarrow -i \sigma_i^x \sigma_i^y \sigma_i^z$$

- Separability of the entanglement entropy (Yao and Qi, PRL105 080501)

$$S_A = S_A^F + S_A^G$$

# Kitaev Model: Gauge Theory

In the Majorana fermion representation, the model is equivalent to a system of free Majorana fermions (matter field) coupled to a static  $\mathbb{Z}_2$  gauge field

- Define the gauge field  $\hat{u}_{ij} = e^{iA_{ij}}$ , then the elementary  $\mathbb{Z}_2$  gauge flux reads

$$e^{i\Phi} = \exp\left(i \oint_{\partial p} A\right) = \prod_{\langle ij \rangle \in \partial p} \hat{u}_{ij} = \prod_{\langle ij \rangle \in \partial p} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} =: W_p$$

- The electric field in the gauge field that flips the gauge field is  $b_i^{\alpha_{ij}} \rightarrow e^{i\pi E_{ij}}$  because  $b_i^{\alpha_{ij}} \hat{u}_{ij} b_i^{\alpha_{ij}} = -\hat{u}_{ij}$
- Gutzwiller projection as a Gauss law constraint:  $D_i = b_i^x b_i^y b_i^z c_i = 1$ . The ground state is

$$\prod_i \left( \frac{1 + D_i}{2} \right) |u\rangle |M\rangle \quad ; \quad \text{Fix the gauge: } \langle u | \hat{u}_{ij} | u \rangle = +1 \forall \langle ij \rangle$$

- Both definitions  $b_i^{\alpha_{ij}} = e^{i\pi E_{ij}}$  and  $b_j^{\alpha_{ij}} = e^{i\pi E_{ij}}$  are valid.
- The Gauss law is a physical constraint and cannot be broken.

# Kitaev Model: Properties of Spin Correlation Functions

- Two-spin correlation functions are short-ranged (Baskaran, Mandal, and Shankar, PRL**98**, 247201)

$$\langle \sigma_i^\mu \sigma_j^\nu \rangle = \langle M | i c_i c_j | M \rangle \underbrace{\langle u | i b_i^\mu b_j^\nu | u \rangle}_{\text{vanishes if } i b_i^\mu b_j^\nu \neq \hat{u}_{ij}}$$

- Multi-spin correlation functions can ONLY be string-like

$$\left\langle \prod_{\langle ij \rangle \in \mathfrak{D}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \right\rangle \equiv \langle \Sigma_{\mathfrak{D}} \rangle$$

- $\mathbb{Z}_2$  algebra of strings

$$\mathfrak{D} \oplus \mathfrak{D}' := \mathfrak{D} \cup \mathfrak{D}' - \mathfrak{D} \cap \mathfrak{D}' \quad : \quad \textit{illustration}$$

- String operators form a projective representation

$$\boxed{\Sigma_{\mathfrak{D}} \Sigma_{\mathfrak{D}'} = \varphi(\mathfrak{D}, \mathfrak{D}') \Sigma_{\mathfrak{D} \oplus \mathfrak{D}'}} \quad ; \quad \varphi(\mathfrak{D}, \mathfrak{D}') = \pm 1, \quad \varphi(\mathfrak{D}, \mathfrak{D}) = (-1)^{|\mathfrak{D}|}$$

# Organization: Equivalence Relation

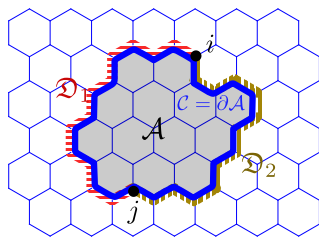
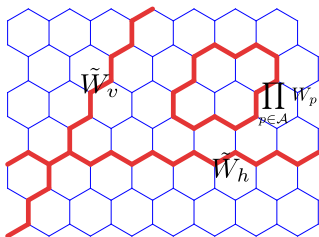
- Exact 1-form symmetry: Wilson loop  $\Sigma_C |\psi_0\rangle = W_C |\psi_0\rangle = |\psi_0\rangle$
- The weights are identical up to a sign if the strings are identical up to a loop  $\mathcal{C}$

$$\langle \Sigma_{\mathfrak{D} \ominus \mathcal{C}} \rangle = \varphi(\mathfrak{D}, \mathcal{C}) \langle \Sigma_{\mathfrak{D}} \Sigma_{\mathcal{C}} \rangle = \varphi(\mathfrak{D}, \mathcal{C}) \langle \Sigma_{\mathfrak{D}} W_{\mathcal{C}} \rangle = \varphi(\mathfrak{D}, \partial A) \langle \Sigma_{\mathfrak{D}} \rangle$$

- Equivalence relation:

$$[\mathfrak{D}] = \{ \mathfrak{D}' : \exists \mathcal{C} \text{ composed by closed loops such that } \mathfrak{D}' \ominus \mathcal{C} = \mathfrak{D} \}$$

- Endpoints perspective:  $\text{EP}(\mathfrak{D}) = \text{EP}(\mathfrak{D}')$  iff  $[\mathfrak{D}] = [\mathfrak{D}'] \Rightarrow$  Emerging fermions at endpoints of strings



# Organization of Density Matrix

- Suppose the observables are a set of  $\mathbb{Z}_2$  traceless Hermitian operators:  $\{\mathcal{O}_n^2 = \mathbb{1}\}$ , then the density matrix can be expanded in terms of these observables

$$\rho \propto \sum_n \text{Tr}(\mathcal{O}_n^\dagger \rho) \mathcal{O}_n = \sum_n \langle \mathcal{O}_n \rangle \mathcal{O}_n$$

- For the spin-1/2 Kitaev model, the observables are Pauli strings.

$$\rho \propto \sum_{\mathfrak{D}} \langle \Sigma_{\mathfrak{D}} \rangle \Sigma_{\mathfrak{D}}$$

- Organization due to the exact 1-form Wilson symmetry:

$$\rho \propto \mathcal{P}^{(p)} \sum_{[\mathfrak{D}]} \langle \Sigma_{\mathfrak{D}} \rangle \Sigma_{\mathfrak{D}} \quad ; \quad \mathcal{P}^{(p)} := \underbrace{\left( \frac{\mathbb{1} + \tilde{W}_h}{2} \right) \left( \frac{\mathbb{1} + \tilde{W}_v}{2} \right)}_{\text{topological projector (global Wilson)}} \underbrace{\prod_p \left( \frac{\mathbb{1} + W_p}{2} \right)}_{\text{flux-free projector}}$$

- Normalize  $\text{Tr} \rho = 1$  to get the final result:

$$\rho = \frac{1}{2^{N-N_p}} \mathcal{P}^{(p)} \sum_{[\mathfrak{D}]} \langle \Sigma_{\mathfrak{D}} \rangle \Sigma_{\mathfrak{D}}$$

# Organization of Reduced Density Matrix

- Following the same spirit to construct the reduced density matrix:

- (i) Hermiticity
- (ii) Ability of reproducing expectation values of observables
- (iii) Normalization condition  $\text{Tr}\rho = 1$

$$\rho_A \propto \sum_{\mathfrak{D} \subset A} \langle \Sigma_{\mathfrak{D}} \rangle \Sigma_{\mathfrak{D}} \propto \mathcal{P}_A^{(p)} \sum_{[\mathfrak{D}] \subset A} \langle \Sigma_{\mathfrak{D}} \rangle \Sigma_{\mathfrak{D}} \xrightarrow{\text{normalize}} \boxed{\frac{1}{2^{N^A - N_p^A}} \mathcal{P}_A^{(p)} \sum_{[\mathfrak{D}] \subset A} \langle \Sigma_{\mathfrak{D}} \rangle \Sigma_{\mathfrak{D}}}$$

- Define *projected string operators*:  $\mathcal{S}_{\mathfrak{D}}^{(A)} := \mathcal{P}_A^{(p)} \Sigma_{\mathfrak{D}}$ . In this way,  $\mathcal{S}_{\mathfrak{D}}^{(A)} = \mathcal{S}_{\mathfrak{D}'}^{(A)}$  if  $\mathfrak{D} \sim \mathfrak{D}'$ .
- Representing density matrices in terms of projected string operators:

$$\boxed{\rho_A = \frac{1}{2^{N^A - N_p^A}} \sum_{[\mathfrak{D}] \subset A} \langle \Sigma_{\mathfrak{D}} \rangle \mathcal{S}_{\mathfrak{D}}^{(A)}}$$

# Organization: Block-Diagonal Structure

$$\mathcal{S}_{\mathcal{D}} = \bigoplus_{n=1}^{2^{L-1}} \mathcal{M}_{\mathcal{D}}^{(n)}$$

- *Majorana representation* of string operators:  $\mathcal{M}_{\mathcal{D}} := \prod_{\langle ij \rangle \in \mathcal{D}} i c_j c_i$ , which satisfies the same algebra

$$\mathcal{M}_{\mathcal{D}} \mathcal{M}_{\mathcal{D}'} = \varphi(\mathcal{D}, \mathcal{D}') \mathcal{M}_{\mathcal{D} \ominus \mathcal{D}'}$$

- Difference in the Hilbert space dimension:

$$\text{Tr}_{A, F} \mathcal{M}_{\emptyset}^{(A)} = 2^{N^A/2} \neq \text{Tr}_A \mathcal{S}_{\emptyset}^{(A)} = 2^{N^A - N_P^A}$$

- Conjecture: projected string operators are *block diagonal*:  $\mathcal{S}_{\mathcal{D}}^{(A)} = \frac{1}{2^{L-1}} \bigoplus_{n=1}^{2^{L-1}} \mathcal{M}_{\mathcal{D}}^{(n)}$

**Observation:** A degenerate entanglement spectrum leads to the separability of the entanglement entropy

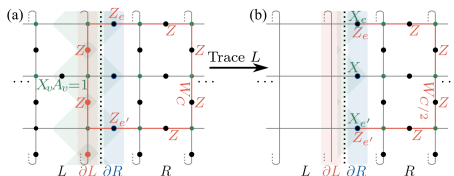
$$\rho = \frac{1}{d} \bigoplus_{n=1}^d \varrho \Rightarrow S^{(n)}(\rho) = \frac{1}{1-n} \ln \left[ \frac{1}{d^{n-1}} \text{Tr}(\varrho^n) \right] = \underbrace{S^{(n)}(\varrho)}_{\text{matter}} + \underbrace{\ln d}_{\text{gauge}}$$

## Questions:

- Can we directly prove that the string operators are block-diagonalized?
- How could they be related to *gauge fields*?

Exact Gauss Law  $\times$  Exact 1-Form Wilson Symmetry:  
–Exact Block-Diagonal Structure

Review the method discovered by a recent work [W.-T. Xu *et al.* PRX **15**, 011001 (2025)]



- Exact Gauss law  $[\rho, X_e] = 0 \ \forall e \in \partial R$ : block-diagonal structure  $\rho = \bigoplus_{\{x\}} \varrho_{\{x\}}$
- Exact 1-form Wilson loop symmetry  $[W_{C/2}, \rho] = 0$  relates different blocks since  $W_{C/2} X_e W_{C/2}^\dagger = -X_e \ \forall e \in \partial R$
- Yielding  $2^{|\partial R|-1}$ -fold degeneracy  $\rho = \frac{1}{2^{|\partial R|-1}} \varrho^{\oplus 2^{|\partial R|-1}}$

# Block-Diagonal Density Matrix in the Kitaev Model

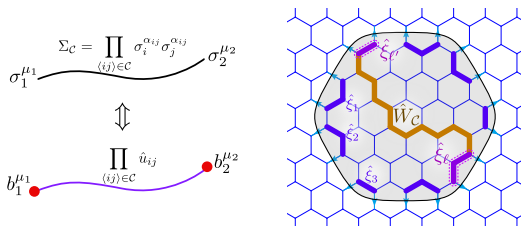
In the Kitaev model:

- Exact Gauss law: From the electric field  $b_i^{\alpha ij} = e^{i\pi E_{ij}}$  at the boundary.
- Exact 1-form Wilson symmetry:  $\langle W_C \rangle = \langle \Sigma_C \rangle = 1$  for any closed loop  $C$ .

But the gauge field here is *fictitious*. To use spin representation (physical):

- Connecting two *deconfined* degrees of freedom to form physical objects
- Use electric fields to define a set of operators  $\hat{\xi}_n := \sigma_n^{\mu_n} \Sigma_{C_n} \sigma_{n+1}^{\mu_{n+1}}$  along the boundary of the entanglement cut.
- Eventually, we obtain  $L = \frac{1}{2} |\partial A|$  independent  $\hat{\xi}$ -operators, and they satisfy

$$[\mathcal{S}_{\mathfrak{D}}^{(A)}, \hat{\xi}_n] = 0 \quad ; \quad \mathfrak{D} \subset A$$



# Block-Diagonal Density Matrix in the Kitaev Model

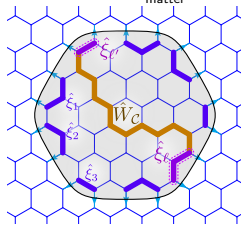
- String operators are block-diagonalized by  $\hat{\xi}$ -operators
- Non-contractible Wilson operators flip  $\hat{\xi}$ -operators, making different blocks identical
- Matching the Hilbert space dimension and considering the identical algebra between  $\mathcal{S}$  and  $\mathcal{M}$

$$\mathcal{S}^{(A)}_{\mathcal{D}} = \frac{1}{2^{L-1}} \bigoplus_{n=1}^{2^{L-1}} \mathcal{M}^{(n)}_{\mathcal{D}}$$

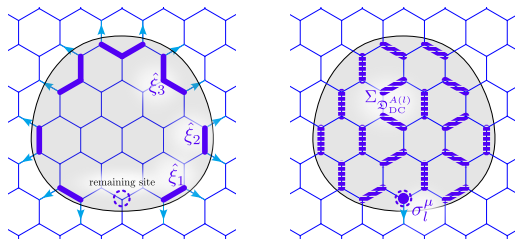
- Reduced density matrix is also block-diagonal: extensive degeneracy of the entanglement spectrum

$$\rho_A = \frac{1}{2^{L-1}} \bigoplus_{n=1}^{2^{L-1}} \rho_A^F \Rightarrow S_A^{(n)} = \frac{1}{1-n} \ln \text{Tr} \rho_A^n = \underbrace{S^{(n)}(\rho_A^F)}_{\text{matter}} + \underbrace{(|\partial A| - 1) \ln 2}_{\text{gauge}}$$

$$\begin{array}{c} \Sigma_C = \prod_{\langle ij \rangle \in C} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \\ \sigma_1^{\mu_1} \text{---} \sigma_2^{\mu_2} \\ \updownarrow \\ \prod_{\langle ij \rangle \in C} \hat{u}_{ij} \\ b_1^{\mu_1} \text{---} b_2^{\mu_2} \end{array}$$



# Odd Number of Sites



- For odd  $N^A$ , there is always an unpaired site at the boundary.
- We can define an additional operator  $\Xi_A$  that also commutes with  $\rho_A$ , yielding an additional 2-fold degeneracy in the entanglement spectrum

$$\hat{\Xi}_A \equiv \sigma_l^\mu \Sigma_{\mathcal{D}_{\text{DC}}}^{A(l)} \rightarrow \left( \prod_{\langle ij \rangle \in \mathcal{D}_{\text{DC}}^{A(l)}} \hat{u}_{ij} \right) \underbrace{\left( b_l^\mu \prod_{i \in A} c_i \right)}_{\text{fermion parity}}$$

# Summary



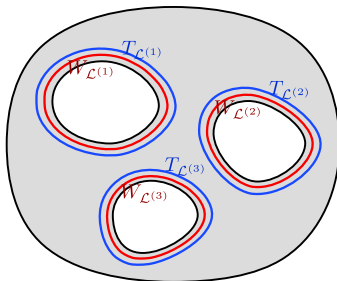
# Unsolved Problems and Outlooks

- To get the REAL topological entanglement entropy, we need an exact 't Hooft symmetry and make it spontaneously broken.
- Higher spin Kitaev model:

# Toric Code Model

- Reduced density matrix of a Kitaev ground state

$$\rho_A = \frac{1}{2^{N^A - N_p^A - N_s^A - 2g}} \prod_{n=1}^g \left( \frac{\mathbb{1} + T_{\mathcal{L}(n)}}{2} \right) \prod_{n=1}^g \left( \frac{\mathbb{1} + W_{\mathcal{L}(n)}}{2} \right) \prod_{p \in A} \left( \frac{\mathbb{1} + P_p}{2} \right) \prod_{s \in A} \left( \frac{\mathbb{1} + S_s}{2} \right)$$

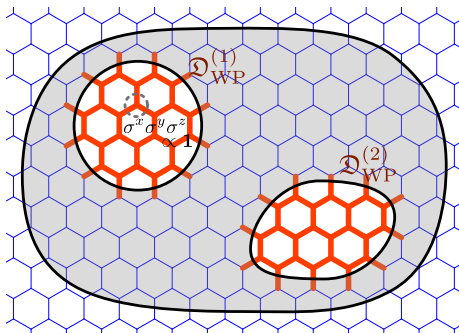


- Entanglement entropy: If the density matrix is proportional to a projector  $\rho_A = C\mathcal{P}$

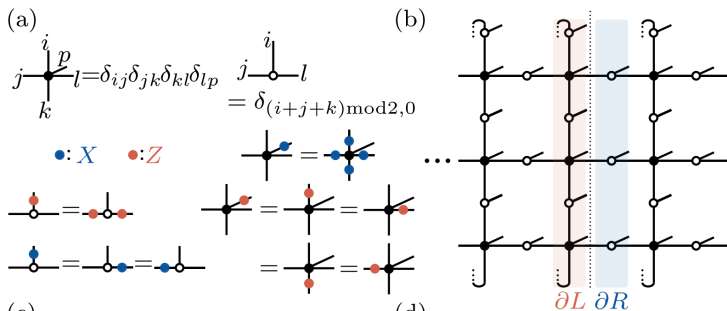
$$S_A^{(n)} = \frac{1}{1-n} \ln \text{Tr} \rho_A^n = \frac{1}{1-n} \ln (C^{n-1} \text{Tr} \rho_A) = -\ln C$$

# Frame Title

- For a multi-connected region with genus number  $g$ , there is a set of *web patch* configurations  $\{\mathfrak{D}_{\text{WP}}^{(1)}, \mathfrak{D}_{\text{WP}}^{(2)}, \dots, \mathfrak{D}_{\text{WP}}^{(g)}\}$  such that  $\text{Tr}_A \left( \Sigma_{\mathfrak{D} \ominus \mathfrak{D}_{\text{WP}}^{(n)}} \right) \neq 0$  if  $\mathfrak{D} \subset A$
- In general, however,  $\Sigma_{\mathfrak{D} \ominus \mathfrak{D}_{\text{WP}}^{(n)}} \neq \Sigma_{\mathfrak{D}} \Sigma_{\mathfrak{D}_{\text{WP}}^{(n)}}$ , thus the reduced density matrix cannot be organized as before.
- The web patch is invalid in the fermion representation:  $\text{Tr}_{A,F} \left( \mathcal{M}_{\mathfrak{D} \ominus \mathfrak{D}_{\text{WP}}^{(n)}} \right) = 0$

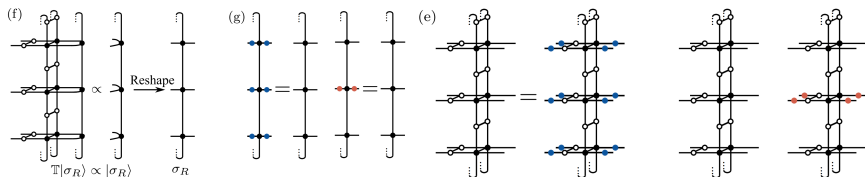


# Contribution 2: Exact 1-Form Wilson



- **Symmetry of the tensor network** derives the **0-form Ising symmetry** and the **1-form Wilson loop symmetry**.

# Contribution 2: Exact 1-Form Wilson



- Transfer matrix of a tensor network gives reduced density matrix (in the virtual space)

$$\mathbb{T}|\sigma_R\rangle \propto |\sigma_R\rangle \Rightarrow |\sigma_R\rangle \rightarrow \sigma_L \Rightarrow \rho \sim \sigma_L^T \sigma_R$$

- Symmetry of the tensor network becomes the symmetry of the reduced density matrix

$$\left( \prod_i X_i \otimes \prod_i X_i \right) \mathbb{T} \left( \prod_i X_i \otimes \prod_i X_i \right) = \mathbb{T} \Rightarrow \left( \prod_i X_i \right) \sigma_{L(R)} \left( \prod_i X_i \right) = \sigma_{L(R)}$$

$$(Z_j \otimes Z_j) \mathbb{T} (Z_j \otimes Z_j) = \mathbb{T} \forall j \quad Z_j \sigma_{L(R)} Z_j = \sigma_{L(R)} \forall j$$

- Therefore, two operators  $\prod_i X_i$  and  $Z_j$  commute with the reduced density matrix, yielding 2-fold degeneracy in the entanglement spectrum.

# Contribution 3: Spontaneous 1-form Symmetry Breaking

- Implementing the 1-form 't Hooft loop symmetry

$$\rho \sim \frac{1 + T_{C_y}}{2} \exp(-H_E)$$

- The 1-form 't Hooft loop symmetry is spontaneously broken:  $T_{C_y} |\psi\rangle \neq |\psi\rangle$ , so the expectation value satisfies the perimeter law:  $\langle T_C \rangle \sim e^{-|C|}$ .
- Entanglement entropy:

$$S_n = \frac{1}{1-n} \ln \text{Tr} \rho^n \stackrel{|\partial R| \rightarrow \infty}{\approx} - \frac{f(1/n) - f(1)}{1/n - 1} |\partial R| \underbrace{- \ln 2}_{\text{topo EE}}$$